Department of Physics and Computer Science

PC141 – Mechanics for Life Sciences
Fall Term 2026

Laboratory Room: 2089, Science Building
Lab Instructor: Dr. Shodiev
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Web: MyLearningSpace

Description

The laboratory component of this course provides students with hands-on experience in mechanics. Through structured experimental investigations, students develop a deeper understanding of physical principles while strengthening experimental technique, data analysis, uncertainty evaluation, and scientific communication skills.

Learning Outcomes

By the end of the lab sequence, students will be able to:

- Set up and conduct experiments in mechanics.
- Collaborate effectively in small groups during investigations.
- Measure and report precision of analog and digital instruments.
- Identify and distinguish between systematic and random errors.
- Estimate, calculate, and interpret uncertainties in measurements and final results.
- Assess whether results agree within uncertainties.

Lab Format

- Students will work in groups of two or three.
- Preparation is essential: read the lab manual and textbook sections and complete any assigned pre-lab tasks before arriving.

- Students must bring the following to every lab session:
    - Lab manual
    - Scientific calculator
    - Notebook
- During lab sessions you will:
    - Demonstrate and discuss your work with the Lab Instructor and IA.
    - Submit in-lab work before leaving.
    - Upload assigned pre-lab and post-lab assignments to MyLearningSpace.

Schedule

- Labs and tutorials begin the **week of May4, 2026**.
- Labs are held every **week**.
- Attendance is taken at every session.
- Punctuality is required: students arriving more than 10 minutes late may not be admitted

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List of Labs

1. **Estimation, Order of Magnitude Calculations, and Bounding** – Introduction to measurement and estimation.
2. **Repeating Measurements** – Precision, accuracy, and repeatability in experiments.
3. **Measuring "g"** – Determining the acceleration due to gravity and examining air resistance effects.
4. **Coefficient of Friction** – Measuring static and kinetic friction coefficients.
5. **Torque and the Principle of Moments** – Investigating torque, rotational equilibrium, and balance.
6. **Moment of Inertia / Lab Evaluation** – Applying rotational dynamics to determine inertia.

Requirements and Policies

Attendance and Makeup Policy

Attendance at all registered laboratory sessions is mandatory.

If you will miss (or have missed) a lab session, email the instructor as soon as possible, preferably before the scheduled session. All follow-up regarding missed labs will be handled in person during your next lab or tutorial session.

Valid reasons for absence and required documentation include:

- Illness: documentation from a licensed medical professional or walk-in clinic
- Varsity athletics travel: official confirmation from Athletics
- Other circumstances: appropriate supporting documentation

Approved make-up labs must be arranged within one week, subject to section capacity and laboratory safety constraints.

Please do not send sensitive medical information by email. Bring required documentation with you to your next lab or tutorial session.

Unexcused absences will result in a grade of zero for the missed laboratory.

Grade Review and Inquiry Policy

- Students must regularly check their grades on MyLearningSpace.
- Questions about grading must be raised within one week of a mark being posted.
- Contact your section Instructional Assistant (IA) first.
- If the issue is not resolved, contact the Lab Coordinator.
- No grade changes will be made after one week.
- All lab and tutorial grades are finalized before the end of the term.
- Requests made after final exams will not be considered.

Lab Notebook

Laboratory work must be completed individually, except that group members may share raw experimental data

Plagiarism Policy

- Work must be completed individually. Groups may share only raw data.
- Examples of plagiarism include:
    - Identical or nearly identical wording in reports.
    - Copying lists, derivations, or calculations from another student.
- Consequences: plagiarism will result in a **grade of zero** on the affected lab and referral to the course instructor and the University's Academic Integrity process
- To avoid plagiarism: write in your own words and from your own understanding.

Digital Devices & Participation Policy

Laboratory sessions are device-free environments.

- The use of cell phones, laptops, tablets, smart watches, or any electronic devices is not permitted during labs. All devices must be turned off and stored away.
- Students must bring a printed copy of the assigned lab manual, a calculator, and a lab notebook. Lab manual for the lab is posted 6-days before the lab day.
  Students who do not bring a printed lab manual (or required materials) will receive a participation grade of -10 for that lab, resulting in a 10% deduction from the total mark for that specific lab session. The lab manual is posted well in advance, and printing or access issues (e.g., printer failure, last-minute printing, location constraints) do not constitute a valid excuse.
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Alternate arrangements will be made in case of illness, holidays, or unforeseen disruptions.